Is electricity production positively associated with GDP/HDI?

- Motivation: the findings inform policy decisions on whether to prioritise electricity infrastructure as a lever for economic development
- Source: Our World in Data
- Time range: 2000–2023
- Key variables and units:
  - Electricity generation (in terawatt-hours)
  - GDP (in international dollars at 2021 prices)
  - HDI (scale from 0 to 1)

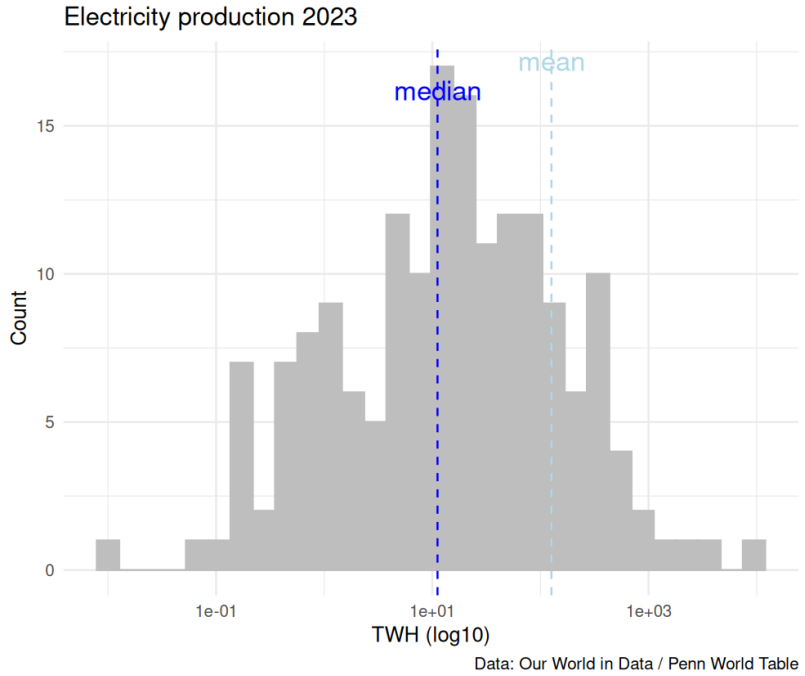
Descriptive Stats

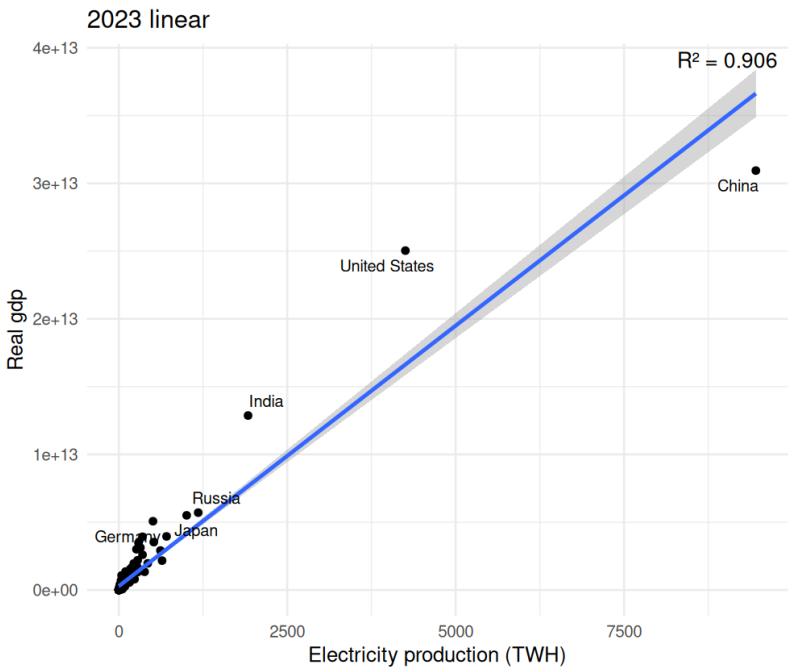
- Electricity Production and GDP → strong right-skew
  - The mean exceeds the median by a factor of 11 for electricity and 7.6 for GDP
  - Most countries have relatively low production but there are a few countries with large production (e.g. China, India, USA), which pull the average upward
  - That is the reason we use the log transformation in later section
- HDI → mild left-skew
  - The mean and median are nearly equal, the distribution is mildly left-skewed
  - In contrast with GDP, HDI is a per-capita index and is on the scale from 0 to 1
  - Most countries cluster at the upper-end of the scale, which limits the scope for variation among developed countries

	Electricity generation	GDP	HDI
Mean	170	940 billion	0.747
Median	15.6	123 billion	0.766
Standard deviation	810	3.27 trillion	0.153
Minimum	0.01	119 million	0.404
Maximum	9,460	30.9 trillion	0.972

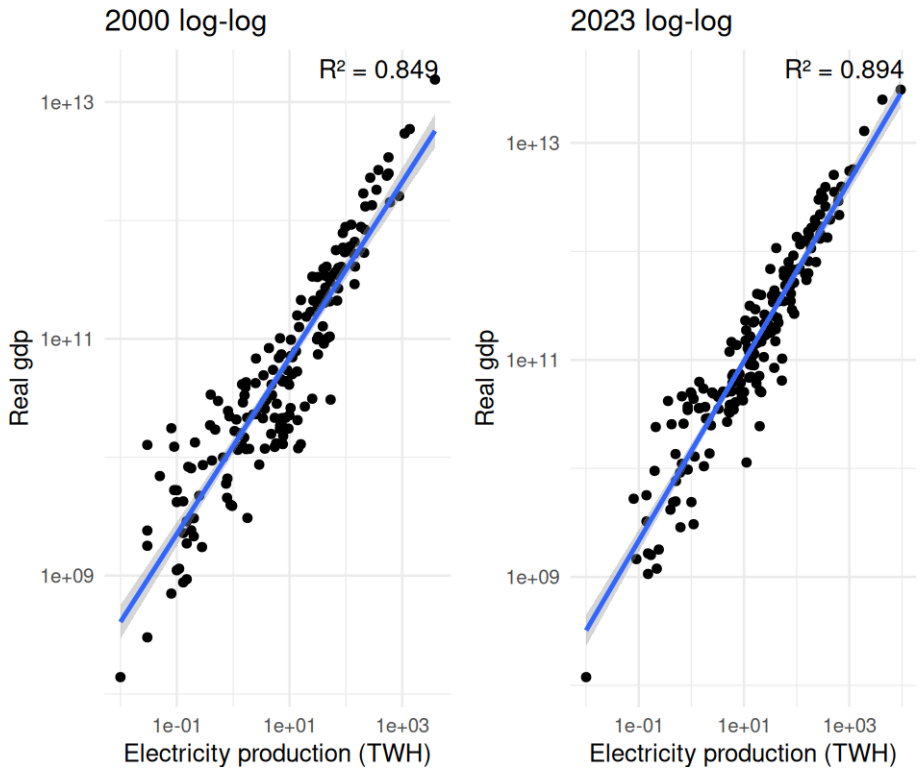
- Most countries produce little electricity, while a few industrial giants (China, USA) push the average up

- The linear scale is misleading because the two outliers (China, USA) have too much influence on the result (the high coefficient of determination is caused only by them)





- On a logarithmic scale, a positive linear relationship is revealed
- The transformation helped linearize the relationship between the variables
- Countries bucking the trend
  - Above trend (high production per given GDP)
    - E.g. Bhutan, Iceland or Paraguay (electricity exporters thanks to hydropower)
    - E.g. Kuwait (resource-rich country)
  - Below trend (low production per given GDP)
    - E.g. Chad, Sierra Leone or Niger (weak infrastructure)
    - E.g. Luxembourg (small economy that imports electricity)



Conclusion & Limitations

- Strong correlation between electricity production, GDP and HDI across all levels of development (confirmed by logarithmic transformation)
- The industrial composition of the country and its natural resources also play a key role
- Higher GDP allows for financing of infrastructure, affordable electricity drives economic growth
- Using total values (without considering the population) may artificially amplify the relationship
- Recommendation for further research: a „per capita“ study to adjust for the effect of country size and more accurately measure efficiency